

09. THE DIFFERENT LAYERS OF THE EYE 1

DURATION : 05:01

STUDY SHEET

COURSE SUMMARY

Analysis of ocular histology (sclera, choroid, ciliary processes, retina) and its functional specificities. Introduction to the principles of iridology as a diagnostic tissue approach. Detailed explanation of photoreceptor and pigment layer activity, illustrating the biochemical mechanisms of visual transduction.

THE DIFFERENT LAYERS OF THE EYE

The **eye** is made up of several layers, often described as superimposed socks.

1. THE SCLERA

The first layer, called the **sclera**, is continuous with the **cornea** at the front. The cornea is a **transparent** structure that allows light to enter the eye. The **limbus** and the white part of the eye, known as the **sclera**, make up this first sock.

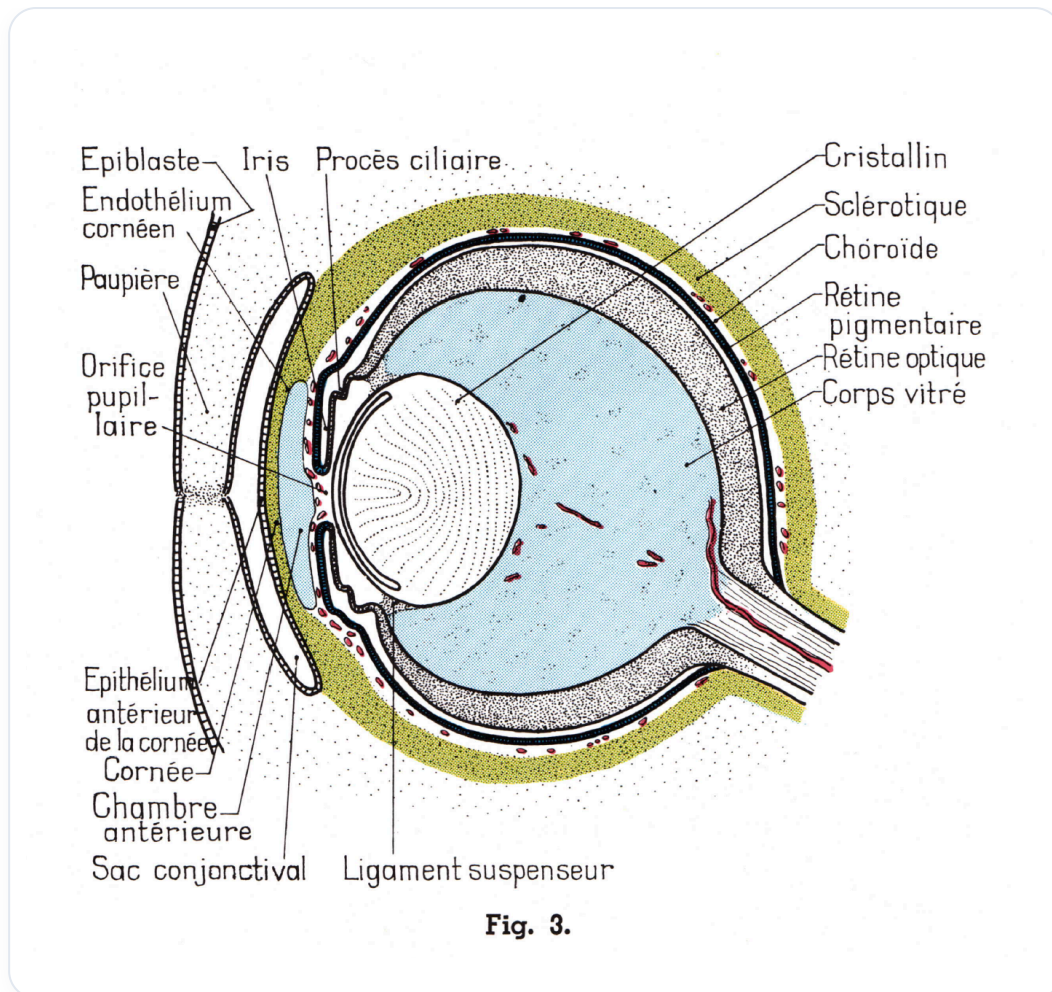


FIG. — General eye diagram

2. THE CHOROID

Upon removing the sclera, the **choroid** is revealed, an extremely **vascularised** layer. The choroid extends forwards with the **iris**. **Iridology** is a practice that involves reading health information through the iris, which can store various elements, such as **sulphur**.

3. THE CILIARY PROCESSES

The **ciliary processes** and the **suspensory muscles of the iris** play a crucial role in the **physiology** of the eye. These structures are essential for the **accommodation** system, allowing the **lens** to change shape so that the image is projected correctly onto the **retina**.

4. THE RETINA

The third sock is the **retina**, which is the deepest nervous tissue and represents an extension of the **brain**. It is formed of two layers: an inner layer and an outer layer. A **detached retina** occurs when the space between these two layers develops.

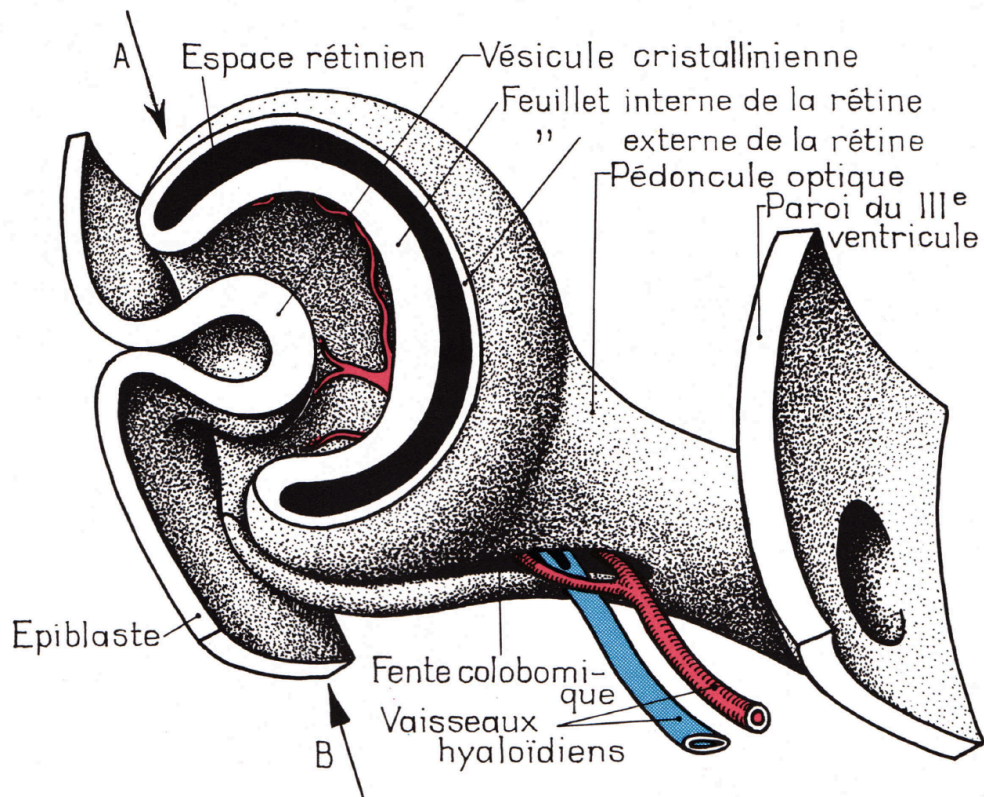


Fig. 3. — Ebauche oculaire humaine. Embryon d'environ 33 jours.

FIG. — Eye tunics, ciliary processes and retina

5. THE PIGMENT LAYER AND PHOTORECEPTORS

The retina also contains a **pigment layer** and **photoreceptors**, which are divided into **cones** and **rods**. These cells are essential for **transduction**, the process by which a photon is converted into electrical energy, thereby enabling visual perception.

CONCLUSION

These different layers of the eye work together to enable vision, each having a specific and crucial role in the functioning of this complex organ.